

Graded Voxels at Fourth Order in Three Dimensions

The Galerkin First-Moment Cube with a Contrast Linear in the Cell

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Abstract

(Written last, from the measured results.)

1 Introduction

2 The graded voxel

A voxel is a cube $V = [-h, h]^3$ of half-width h centred at $\mathbf{x}_c$, with local coordinates $\boldsymbol{\xi} = (\mathbf{x} - \mathbf{x}_c)/h \in [-1, 1]^3$ in the axis order (z, x, y) . The field in it is the nine-component state $\boldsymbol{\psi} = (\mathbf{u}, \boldsymbol{\varepsilon})$: the displacement $\mathbf{u}$ and the engineering strain $\boldsymbol{\varepsilon}$ in Voigt order $(zz, xx, yy, 2xy, 2zy, 2zx)$. The strain is an *independent* unknown, not the gradient of $\mathbf{u}$. A basis whose strain is derived from its displacement leaves a spurious source at every face between cells, because each cell's projected displacement jumps across the face; in a layer that scheme converges at second or third order however rich the basis [?]. The quadratic 27-mode cube T -matrix is of that kind and is not considered further.

The state is expanded in the Legendre polynomials of degree at most one,

$$\boldsymbol{\psi}(\mathbf{x}) = \sum_{a=0}^3 \boldsymbol{\psi}_a L_a(\boldsymbol{\xi}), \quad L_0 = 1, \quad L_i = \xi_{i-1} \ (i = 1, 2, 3), \quad (1)$$

36 weights per cell. The contrast operator $\boldsymbol{\Delta} = \text{diag}(\omega^2 \Delta \rho \mathbf{I}_3, \Delta \mathbf{C})$, with ω the angular frequency, $\Delta \rho$ the density contrast and $\Delta \mathbf{C}$ the 6×6 Voigt stiffness contrast, is linear in the cell,

$$\boldsymbol{\Delta}(\boldsymbol{\xi}) = \boldsymbol{\Delta}_0 + \sum_{i=1}^3 \boldsymbol{\Delta}_i \xi_{i-1}, \quad (2)$$

its coefficients the L^2 projection of the medium onto $\{1, \xi_i\}$. The source density $\boldsymbol{\Delta} \boldsymbol{\psi}$ is then a polynomial of degree at most two. Writing m_c ($c = 0, \dots, 9$) for the ten monomials of degree at most two,

$$\boldsymbol{\Delta}(\boldsymbol{\xi}) L_b(\boldsymbol{\xi}) = \sum_{c=0}^9 \mathbf{E}_{cb} m_c(\boldsymbol{\xi}), \quad (3)$$

where $\mathbf{E}_{cb}$ collects the 9×9 coefficients: the Legendre linearisation of the one-dimensional construction [?], with no new kernel object. The Lippmann–Schwinger equation $\psi = \psi^0 + \int \mathbf{P}(\mathbf{x} - \mathbf{x}') \Delta \psi d^3 x'$, with ψ^0 the incident state and $\mathbf{P}$ the 9×9 point propagator $[[\mathbf{G}, \mathbf{C}], [\mathbf{H}, \mathbf{S}]]$ built from the elastodynamic Green’s tensor and its derivatives, is tested with each L_a over each cell (Galerkin). Cells m and n couple through the blocks

$$\mathbf{K}_{ac}(\mathbf{R}) = \int_{V_m} \int_{V_n} L_a(\xi) \mathbf{P}(\mathbf{x} - \mathbf{x}') m_c(\xi') d^3 x d^3 x', \quad \mathbf{R} = \mathbf{x}_m - \mathbf{x}_n, \quad (4)$$

and the discrete system is

$$\sum_b \mathbf{M}_{ab} \psi_{m,b} - \sum_n \sum_{b,c} \mathbf{K}_{ac}(\mathbf{R}_{mn}) \mathbf{E}_{cb}^{(n)} \psi_{n,b} = \langle L_a, \psi^0 \rangle_m, \quad (5)$$

with $\mathbf{M}_{ab} = \langle L_a, L_b \rangle \mathbf{I}_9$ the Gram matrix, $\text{diag}(8h^3, 8h^3/3, 8h^3/3, 8h^3/3)$.

3 The single site

For one isolated cell the system closes on the self block $\mathbf{K}(0)$. With the source moments $\sigma_a = \langle L_a, \Delta \psi \rangle = \sum_b \mathbf{F}_{ab} \psi_b$, $\mathbf{F}_{ab} = \sum_c \langle L_a, m_c \rangle \mathbf{E}_{cb}$, the graded single site is

$$\mathbf{T}_{36} = \mathbf{F} (\mathbf{M} - \mathbf{K}(0) \mathbf{E})^{-1} \mathbf{M}, \quad (6)$$

which maps the incident field’s Legendre coefficients to the source moments: force and stress means and their first moments. The first moment of the force, which a nine-component composite cannot hold, is part of the object.

Parity. Under inversion $\mathbf{x} \rightarrow -\mathbf{x}$ the displacement is odd, the strain even and ξ_i odd. With $g = 0$ (no gradient) $\mathbf{T}_{36}$ therefore splits exactly into a 15-dimensional set (strain $\times 1$, displacement $\times \xi$) and a 21-dimensional set (displacement $\times 1$, strain $\times \xi$); the coupling between them is 5×10^{-18} of the largest entry. A gradient is odd about the centre and couples the two sets linearly: $0.325 gh$ of the largest entry, for a gradient g along one axis, at $gh = 0.05, 0.1$ and 0.2 . An axis-aligned gradient leaves the symmetry of rotations by a quarter turn about its axis, which $\mathbf{T}_{36}$ satisfies to 10^{-11} .

Reciprocity. The propagator satisfies $\mathbf{W}\mathbf{P}(\mathbf{r})$ symmetric at every $\mathbf{r}$, with $\mathbf{W} = \text{diag}(\mathbf{I}_6, \frac{1}{2}\mathbf{I}_3)$, and $\mathbf{P}(-\mathbf{r}) = \mathbf{\Pi}\mathbf{P}(\mathbf{r})\mathbf{\Pi}$ with $\mathbf{\Pi} = \text{diag}(-\mathbf{I}_3, \mathbf{I}_6)$. Every block between test functions is therefore $\mathbf{W}$ -symmetric, $(\mathbf{W}\mathbf{K}_{ab})^\top = \mathbf{W}\mathbf{K}_{ab}$, and the self block obeys $\mathbf{K}_{ba}(0) = \mathbf{\Pi}\mathbf{K}_{ab}(0)\mathbf{\Pi}$ (exchange the two points of the cell). Both hold to 10^{-11} .

The Born limit. Without a gradient $\mathbf{T}_{36}$ is exact at first order in the contrast for any incident field: the source moments are $\Delta_0 \langle L_a, \psi^0 \rangle$. With a gradient the first-order source $\langle L_a, (\mathbf{g} \cdot \xi) \psi^0 \rangle$ needs the incident field’s *quadratic* moments, which the degree-one input drops. The error is $O(\Delta gh (kh)^2)$, with k the incident wavenumber: measured slopes 2.0001 and 2.0004 for

$kh = 0.025 \rightarrow 0.05 \rightarrow 0.1$, where it reaches 8.9×10^{-5} at $gh = 0.2$. It is the single site's view of the rule $\min(2p + 2, 2r + 2)$ for a field of degree p and a contrast of degree r [?].

An isolated cube has an error floor that refinement in k does not remove. In the static limit a cube of contrast $\Delta\mathbf{C}$ under a uniform remote strain has the internal strain $\boldsymbol{\varepsilon}(\mathbf{x}) = \boldsymbol{\varepsilon}^0 + \int_V \partial\partial\mathbf{G}(\mathbf{x} - \mathbf{x}') \Delta\mathbf{C} \boldsymbol{\varepsilon}(\mathbf{x}') d^3x'$. The self-interaction $\mathbf{S}(\mathbf{x}) = \int_V \partial\partial\mathbf{G}(\mathbf{x} - \mathbf{x}') d^3x'$ is constant inside an ellipsoid and only there [?]; in a cube it concentrates towards the edges and corners. A closure that keeps averages is exact through second order in the contrast, since the second-order source needs only $\int_V \mathbf{S}$, and first errs at third order, by $\Delta\mathbf{C}^3 V \text{Var}(\mathbf{S})$; a collocation closure, which uses $\mathbf{S}(0)$, errs already at second order. Neither error depends on k . It is not what a tiled medium sees: between identical cells the faces are not physical, the exact field is smooth there, and the concentration does not arise. The order of convergence is a property of the tiled medium, the closure error a property of the isolated cell; this paper measures the former (§5).

4 The coupling between cells

The point propagator. $\mathbf{P}$ is built from $\mathbf{G}_{ij} = \frac{1}{4\pi\mu} [\delta_{ij}g_S(r) + k_S^{-2}\partial_i\partial_j(g_S - g_P)(r)]$, $g_k = e^{ikr}/r$, with μ the background shear modulus and k_P, k_S the P and S wavenumbers. Every derivative of a radial function f is a combination of $F_q = (r^{-1}d/dr)^q f$. Near the origin the closed form cancels catastrophically (the k_S^{-2} difference), so for $k_S r \leq \frac{1}{2}$ the F_q are summed term by term from the power series of g_k . This kernel equals the package's closed form to 10^{-11} where both are accurate, and its static part is the Kelvin tensor to 10^{-13} . Its static (Kelvin) part is the $t = 0$ term of the δ_{ij} series and the $t = 2$ term of the derivative series, $\frac{1}{4\pi\mu}[\delta_{ij}r^{-1} + b_2\partial_i\partial_jr]$ with $b_2 = -\frac{1}{2}(1 - \beta^2/\alpha^2)$; the dynamic remainder is at most $1/r$.

The s-form. With $\mathbf{x} = \mathbf{x}_m + \mathbf{u}$ and $\mathbf{x}' = \mathbf{x}_n + \mathbf{u}'$ the kernel sees $\mathbf{R} + \mathbf{s}$, $\mathbf{s} = \mathbf{u} - \mathbf{u}' \in [-2h, 2h]^3$, and Eq. (4) becomes

$$\mathbf{K}_{ac}(\mathbf{R}) = \int W_{ac}(\mathbf{s}) \mathbf{P}(\mathbf{R} + \mathbf{s}) d^3s, \quad W_{ac}(\mathbf{s}) = \prod_i w_{e_{a,i}e_{c,i}}(s_i), \quad (7)$$

with the one-dimensional cross-correlation $w_{\alpha\beta}(s) = h \int v^\alpha (v - s/h)^\beta dv$ over $v \in [\max(-1, s/h - 1), \min(1, s/h + 1)]$: continuous, a polynomial on $[-2h, 0]$ and on $[0, 2h]$, zero at $\pm 2h$, with kinks at 0 and $\pm 2h$. For cells that do not touch the kernel is smooth on each of the eight pieces and a three-dimensional Gauss rule suffices ($8n^3$ kernel evaluations against n^6 for the double integral). It agrees with six-dimensional Gauss quadrature of Eq. (4) to 10^{-9} one cell beyond contact.

The self cell and its 26 neighbours. The static part is a sum of terms $\partial^l r^m$ ($m = -1$ with $|\boldsymbol{\iota}| \leq 2$, $m = 1$ with $|\boldsymbol{\iota}| \leq 4$), distributions at the origin. Moving $k = \max(0, |\boldsymbol{\iota}| - (m + 1)) \leq 2$ derivatives onto W leaves a kernel at most $1/r$: $\int W \partial^l f = (-1)^k \int (\partial^{\text{moved}} W) \partial^{\text{rest}} f$. Two derivatives on one axis turn the kinks of w into plane deltas $J_b \delta(s - bh)$, $b \in \{-2, 0, 2\}$, J_b the jump of w' at b . Every piece is then a box or a rectangle whose corners lie on the breakpoints, so the singular

point $\mathbf{s} = -\mathbf{R}$ is either a vertex, integrated with Duffy's pyramids, or at distance $2h$ or more. The Eshelby delta is not added by hand; it is carried by the plane deltas. The construction reproduces the Eshelby sum rule $\sum_p \iint_{V \times V} \partial_p \partial_p r^{-1} = -4\pi V$ and the cube's Coulomb self-energy

$$\iint_{V \times V} \frac{d^3x d^3x'}{|\mathbf{x} - \mathbf{x}'|} = 2 \left[\frac{1}{5} (1 + \sqrt{2} - 2\sqrt{3}) - \frac{\pi}{3} + \ln((1 + \sqrt{2})(2 + \sqrt{3})) \right] (2h)^5 \quad (8)$$

to 10^{-11} , converges in the Gauss order to 10^{-11} , and satisfies the reciprocity and exchange laws of §3.

An independent construction. Integration by parts can move the derivatives onto the cells' polynomials instead of onto W : one derivative onto each cell, each move either differentiating the polynomial or producing a face term, which leaves $1/r$ kernels over cell×cell, face×cell and face×face, all absolutely convergent. Built in Mathematica on its own (`GradedVoxel_WeakKernel.wl`), this reproduces the sum rule and the Coulomb energy, and, as the source cell shrinks to a point, the distributional single average of the moment engine [?]; all to 10^{-14} or better.

5 The smoothly graded sphere

The body and its exact solution. A sphere of radius $a = 10$ m has a homogeneous core $r < a/2$ with the contrast $\Delta\lambda = 2$ GPa, $\Delta\mu = 1$ GPa, $\Delta\rho = 100 \text{ kg m}^{-3}$ in a background $\alpha = 5 \text{ km s}^{-1}$, $\beta = 3 \text{ km s}^{-1}$, $\rho = 2500 \text{ kg m}^{-3}$. Across the shell $a/2 < r < a$ the contrast is multiplied by the smoothstep $10x^3 - 15x^4 + 6x^5$, $x = (a - r)/(a/2)$, so it vanishes like $(a - r)^3$ at the surface: there is no staircase. Its per-order T -matrices are exact (Takeuchi–Saito integration, two implementations agreeing to 2×10^{-11}) [?]. The incident wave is a P wave along z , at $k_s a = 0.5$ and 1 .

Arms and measurement. On the same grid of n^3 cells across the bounding cube: the collocation uniform-field voxel with each cell at the profile's centre value (T_9); the Galerkin uniform-field voxel ($p = r = 0$); and the graded first-moment voxel ($p = r = 1$). A Galerkin cell carries the exact projection of its contrast, so every cell overlapping the sphere is kept: keeping only cells whose centre lies inside drops contrast from partly filled shell cells, and the uniform Galerkin voxel then converges at order 1.07 instead of 2. The error is the largest far-field difference from the exact solution over nine angles, relative to its peak, at 5×10^8 radii; at 5×10^4 radii the near-field terms that the asymptotic far field omits are 1.4×10^{-4} , larger than the graded voxel's own error.

The discretisation alone. At a contrast 10^{-6} of the above, multiple scattering is negligible and the error is that of the discretisation (Table 1). The graded voxel converges at fourth order, with apparent orders 4.01 and 4.13 on the finest pair; both uniform-field voxels are of second order. At $n = 4$ a cell is as wide as the graded shell and the orders there are not asymptotic.

n	$k_S a = 0.5$			$k_S a = 1$		
	T_9	$p = r = 0$	$p = r = 1$	T_9	$p = r = 0$	$p = r = 1$
4	1.04×10^{-1}	8.85×10^{-3}	7.53×10^{-5}	1.16×10^{-1}	3.45×10^{-2}	2.32×10^{-4}
6	4.71×10^{-3}	3.61×10^{-3}	5.03×10^{-6}	8.26×10^{-3}	1.42×10^{-2}	9.14×10^{-5}
8	2.71×10^{-3}	2.06×10^{-3}	1.59×10^{-6}	5.17×10^{-3}	8.15×10^{-3}	2.78×10^{-5}
order 6 $\rightarrow$ 8	1.92	1.96	4.01	1.63	1.93	4.13

Table 1. Weak contrast (10^{-6}): the discretisation error of the far field.

Full contrast. With multiple scattering (Table 2) the graded voxel's error falls to 6.3×10^{-5} and 7.6×10^{-5} at $n = 8$, 42 and 66 times below the collocation voxel on the same grid; at $k_S a = 1$ it is below the collocation voxel at $n = 24$ (4.2×10^{-4} , with 64,872 unknowns against 14,688). Its apparent order between $n = 6$ and 8 is 3.35 and 3.40, and rising. The part of the error that multiple scattering adds is carried by the second-order Born term T_2 (one application of the coupling): computed on its own, which needs no solve and so reaches finer grids, its apparent order rises through 2.85, 3.40, 3.52 and 3.65 from $n = 4$ to 12 at $k_S a = 0.5$ (errors 4.1×10^{-2} to 1.1×10^{-3} of T_2), consistent with fourth order approached slowly, as $h^4 \log(1/h)$ would be; a third-order component would make it fall instead. The dense solve used here is limited by memory to $n = 8$; an FFT matrix-vector product would take the full solve further.

n	$k_S a = 0.5$			$k_S a = 1$		
	T_9	$p = r = 0$	$p = r = 1$	T_9	$p = r = 0$	$p = r = 1$
4	1.03×10^{-1}	7.70×10^{-3}	5.47×10^{-4}	1.15×10^{-1}	3.24×10^{-2}	6.34×10^{-4}
6	4.66×10^{-3}	3.04×10^{-3}	1.65×10^{-4}	8.15×10^{-3}	1.32×10^{-2}	2.02×10^{-4}
8	2.64×10^{-3}	1.68×10^{-3}	6.28×10^{-5}	5.01×10^{-3}	7.52×10^{-3}	7.60×10^{-5}
order 6 $\rightarrow$ 8	1.98	2.05	3.35	1.70	1.95	3.40

Table 2. Full contrast: the far-field error of the three voxels.

A convention the Born limit cannot see. The shear entries of the source are engineering stress $2\sigma_{pq}$: the contrast operator maps the engineering strain γ_{pq} to $2\Delta\mu\gamma_{pq}$, and the kernel's shear columns carry a factor $\frac{1}{2}$. The far field must take the tensor stress σ_{pq} from them, half the entry. Radiating the entry itself doubles the shear stress, and a P wave along a cell axis has no shear stress at Born order, so the weak-contrast test cannot see it. It appears at second order, through the density-shear cross terms, as an error in T_2 that does not fall with refinement (4.65% and 4.05% at $n = 4$ and 6, against 4.22% and 1.31% when the far field agrees with the kernel for every source component).

6 Discussion

7 Conclusions

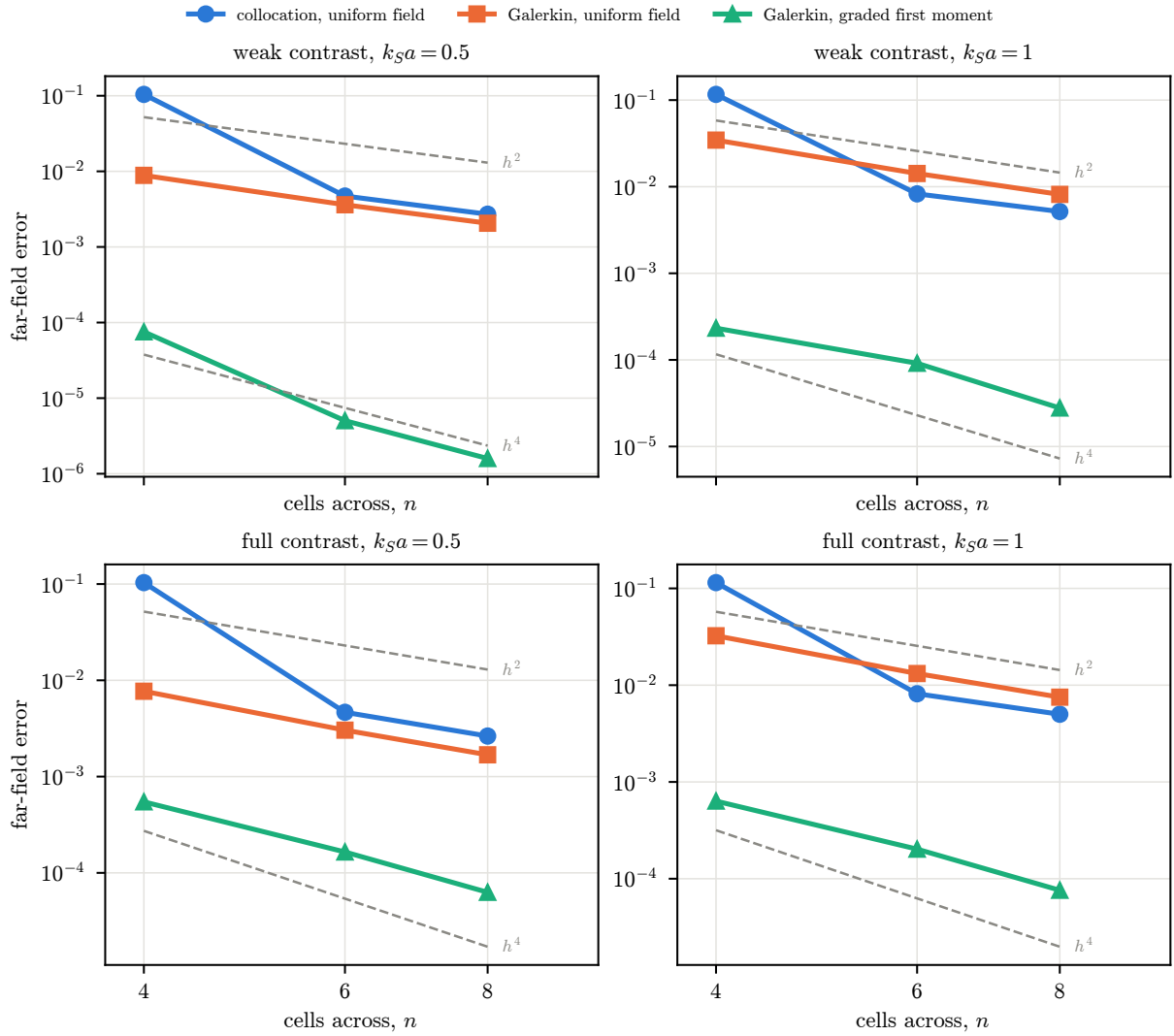


Figure 1. Far-field error against the number of cells across the sphere, for the collocation and Galerkin uniform-field voxels and the graded first-moment voxel; dashed guides show h^2 and h^4 . Top: weak contrast (the discretisation alone). Bottom: full contrast.